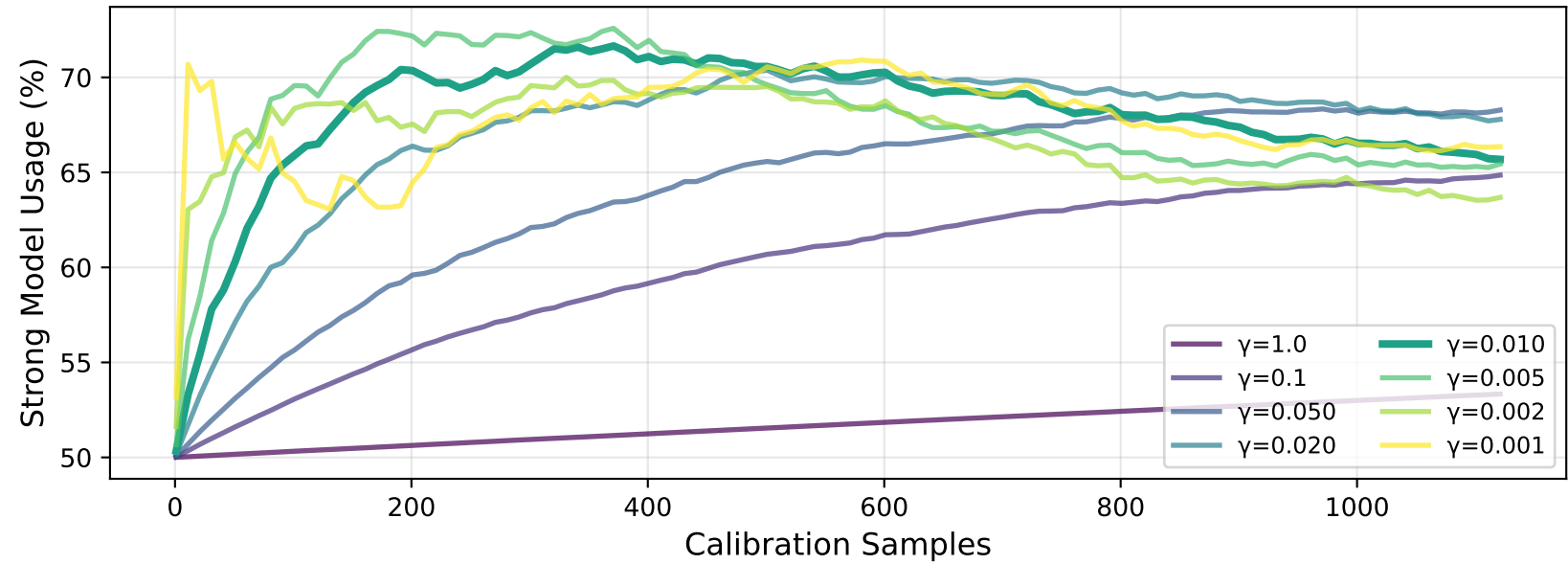
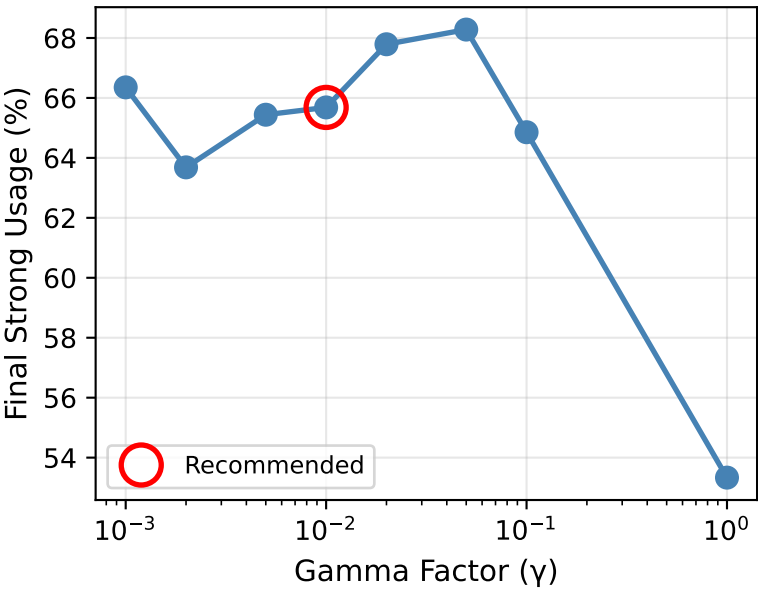


**Figure 3: Optimal Gamma Calibration Analysis**  
**Finding the Balance Between Warmup Priors and Calibration Data**

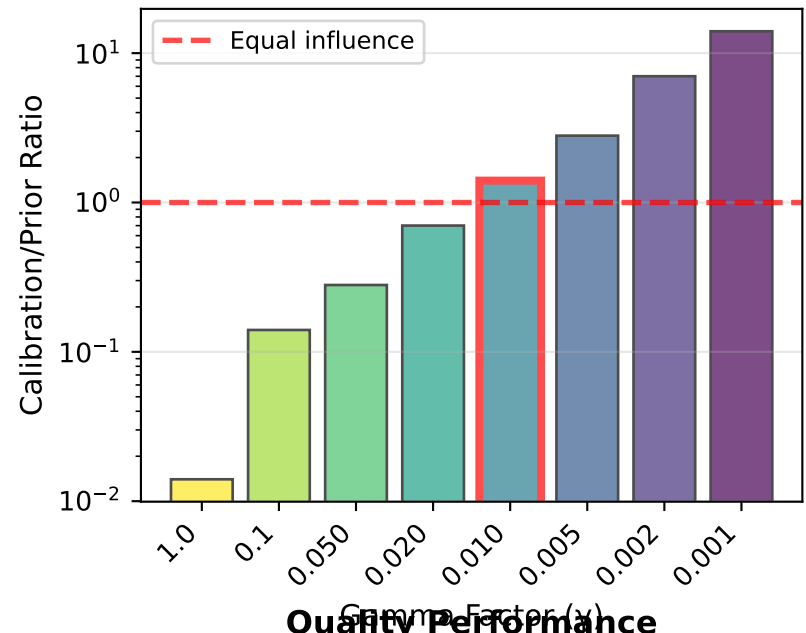
**Policy Adaptation Curves by Gamma**



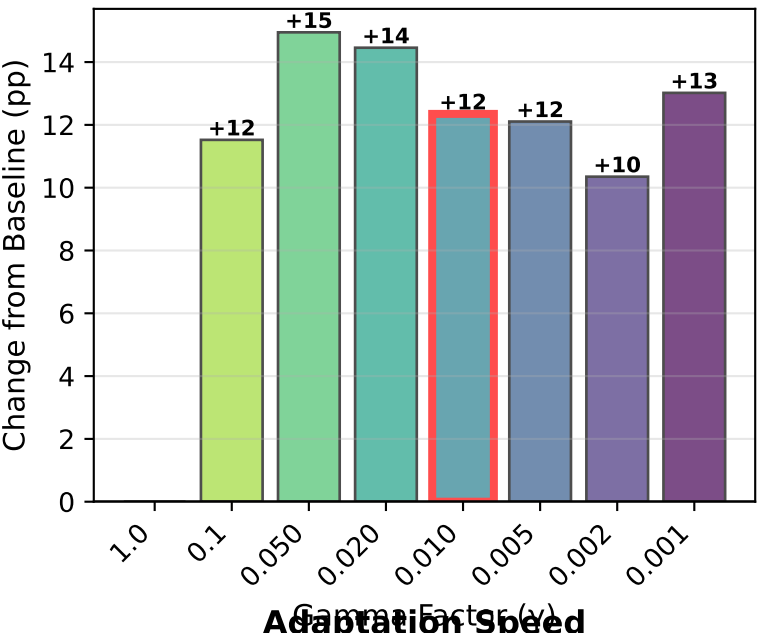
**Prior Weakening Effect**



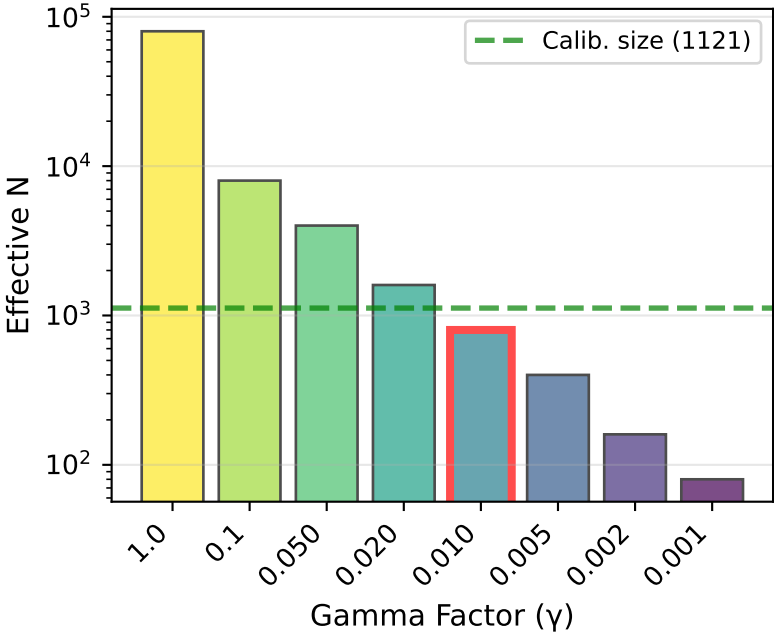
**Influence Balance**



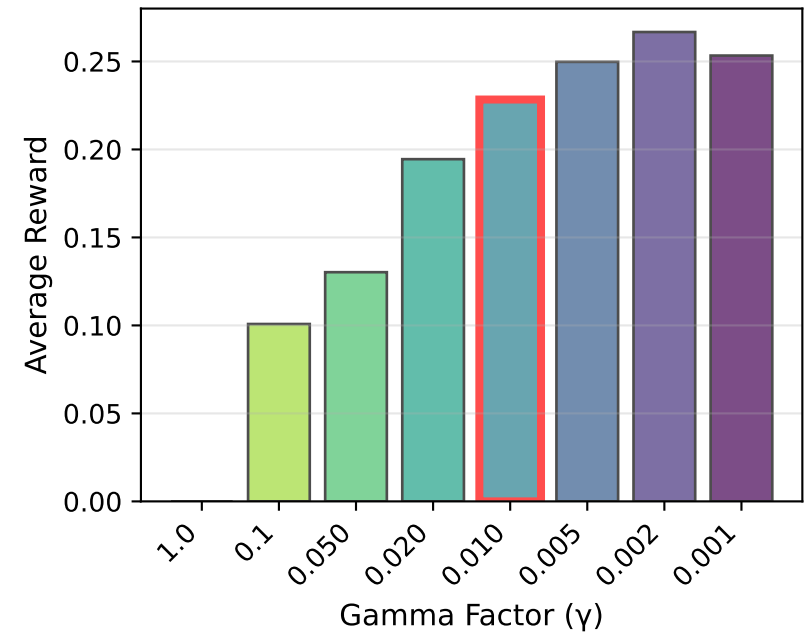
**Adaptation Magnitude**



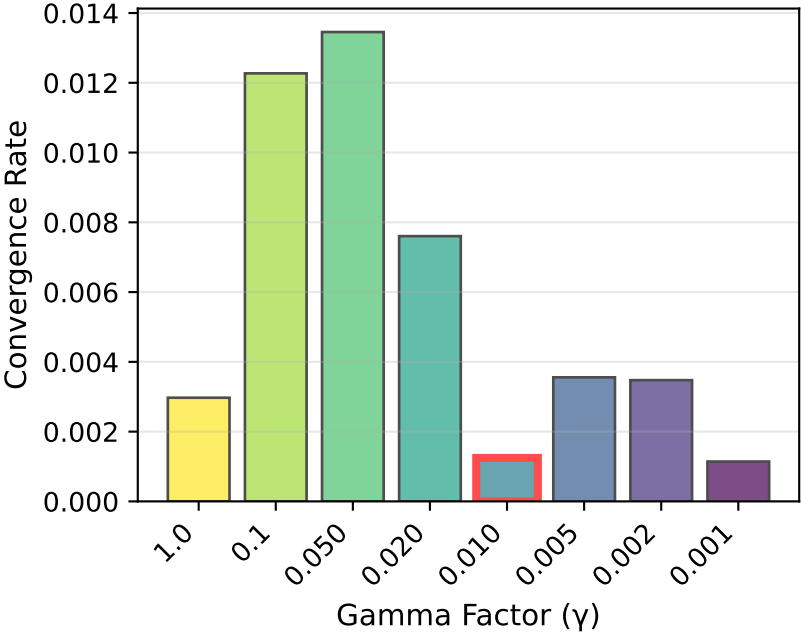
**Prior Strength**



**Quality Performance**



**Adaptation Speed**



**OPTIMAL GAMMA ANALYSIS**

**Dataset:**

- Calibration: 1,121 samples
- Warmup: 80,000 samples
- Models: Mixtral 8x7B Instruct vs Gpt 4 Turbo

Recommended:  $\gamma = 0.010$

**Key Metrics:**

- Strong usage: 65.7%
- Avg reward: 0.2284
- Calib/Prior: 1.401
- Eff. N: 800

**Baseline ( $\gamma=1.0$ ):**

- Strong usage: 53.3%
- Change: +12.4 pp

**Next Steps:**

1. Use  $\gamma=0.010$  for calibration
2. Run `calibrate_router.py`
3. Evaluate on holdout set